

PROJECT PROPOSAL • EMBEDDED SYSTEMS

LASER HARP

A Touchless, Gesture-Controlled Musical Instrument Using the ATmega32

TARGET MCU

Microchip ATmega32 (8-bit AVR)

INTERFACE TYPE

Optical ADC Sensing + PWM Audio

PROJECT TEAM

Md. Sium

ID: 2205153

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Project Motivation & Objectives

Problem Statement

Conventional musical instruments require physical contact and tactile mastery. Touchless optical interaction offers an intuitive, friction-free alternative for expressive sound generation.

Why It Matters

Fuses optics, real-time embedded sensing, and signal processing. Practical applications span adaptive instruments for motor-accessibility needs and interactive art installations.

Embedded Systems Value

Combines multi-channel ADC sampling, debouncing, hardware timers, and communication interfaces into an interactive, real-time physical system rather than an abstract sensor lab.

Main Project Objectives

- 1 Reliable Beam Interruption Sensing**
Real-time ADC light detection across 6-8 laser strings.
- 2 Low-Latency Tone Generation**
Map beam cuts to musical notes with sub-100 ms response time.
- 3 Audible Sound Output**
Produce clean frequency tones via PWM-driven buzzer/speaker.
- 4 Synchronized Visual Feedback**
Trigger visual LED confirmation instantly upon note activation.
- 5 Rigid Optical Frame**
Construct a stable enclosure ensuring exact laser-to-sensor alignment during play.

Hardware & Software Stack

Hardware Components

Component	Qty / Spec	Functional Purpose
ATmega32 MCU	1x DIP-40	Core MCU; handles ADC, logic & tone PWM
Laser Diodes	6-8x 5mW 650nm	Emits optical "strings" across open air
LDR Sensors	6-8x Photocell	Converts light intensity to voltage level
Resistors	6-8x 10kΩ	Voltage divider network for ADC inputs
Audio Output	1x Passive Buzzer	Generates audible square wave tones
LED Strip	1x WS2812 (Opt.)	Visual note confirmation & feedback
Power & Frame	5V DC Regulated	Acrylic/wood rigid optical alignment frame

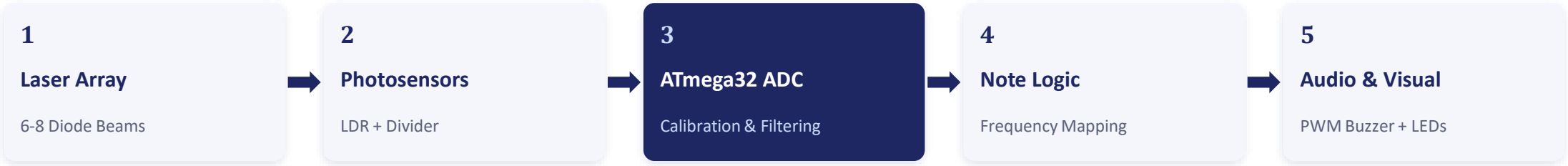
Software & Toolchain

- **Atmel / Microchip Studio**
C/C++ firmware development and debugging platform.
- **AVR-GCC & avrdude**
Cross-compiler toolchain and ISP flashing via USBASP.
- **Proteus Design Suite**
Schematic simulation and ADC logic verification pre-build.
- **Fritzing / KiCad**
Circuit design and wiring schematic documentation.

Bare-Metal AVR C Strategy

Firmware uses direct register manipulation for maximum execution speed and microsecond-level timing control.

System Architecture & Block Diagram



Subsystem Operations

- 1

Optical Emitters

Lasers emit parallel 650nm red beams creating touchless strings.
- 2

Light Sensing

Hand interruption drops lux on the LDR, shifting voltage from ~4.5V to <1.5V.
- 3

ATmega32 Processing

Multi-channel ADC reads levels, compares against dynamic thresholds, and runs debounce filters.
- 4

Output Stage

Timer1 PWM drives the piezo speaker at precise musical frequencies (e.g., C4=261Hz, D4=293Hz).

Central Role of the ATmega32

100% STANDALONE REAL-TIME CONTROLLER

The ATmega32 handles all time-critical sensing, signal processing, and sound generation independently, without external assistance.

Real-Time Tasks Managed by the ATmega32

- 1 Multi-Channel ADC Sampling**
Sequentially polls 6-8 analog photo-sensor channels in round-robin sequence.
- 2 Startup Calibration**
Reads ambient light level on boot to establish relative trigger thresholds per beam.
- 3 Debounce & Glitch Filtering**
Rejects optical noise and transient breaks through a hysteresis algorithm.
- 4 Frequency Mapping & PWM**
Calculates timer reload values to output accurate pitch frequencies via Timer1 PWM.
- 5 Visual Synchronization**
Toggles GPIO pins synchronously to illuminate note-indicator LEDs.

ATmega32 Microcontroller

High-performance 8-bit AVR RISC MCU running at 16 MHz clock speed.

Peripheral Utilization Summary

- ADC (multi-channel)
- Timer1 (PWM audio)
- GPIO (LED outputs)
- UART (ESP32 link)

Implementation Plan & Milestones

Update 1: Early Milestone

- Finalize circuit schematics and acquire optical/passive components.
- Build a single laser-LDR sensor pair test rig on breadboard.
- Program ATmega32 ADC sampling, ambient baseline calibration & threshold logic.
- Verify low-latency tone triggering on a single beam-to-note setup.

Final Stage: Complete System

- Scale optical array to the full 6-8 laser beams with individual LDR channels.
- Assemble the physical wooden/acrylic frame for rigid beam alignment.
- Implement full musical scale mapping & LED feedback synchronization.
- Conduct live presentation demo calibration under ambient room lighting.

Milestone Sequence: Sensor Calibration → Multi-Beam Scaling → Alignment Frame → Final Tuning

Member Responsibilities & Key Dependencies

Md. Sium

ID: 2205153

Circuit & Sensors

Hardware schematic design, voltage divider setup, and sensor response testing.

Rawnak Mubtasim Rafid

ID: 2205154

ATmega32 Firmware

ADC driver, auto-calibration code, debounce algorithms, and interrupt handling.

Muttakin Ahmed

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Enclosure, Audio & Integration

Mechanical frame fabrication, laser alignment, tone-scale mapping, and system integration/testing.

EXTENSION

Secondary Controller Integration (ESP32)

UART Interface & Data Isolation Architecture



Extended ESP32 Features (Optional)

- 1 DFPlayer Mini Trigger**
Triggers high-fidelity audio samples (polyphonic harp / synth) from an SD card.
- 2 WS2812 RGB Animations**
Drives multi-LED color-wave animations synced with the played note frequencies.
- 3 Wireless Connectivity**
Optional Bluetooth / Wi-Fi output for MIDI streaming to a DAW.

Strict Functional Decoupling

Course Evaluation Safeguard

The ESP32 is strictly an optional peripheral extension. If disconnected entirely, the ATmega32 continues to detect beams and produce audible musical output via its local speaker.

Testing Protocol & Success Criteria

Testing Methodology

- 1

Single-Beam Bench Test
Verify voltage swing and calibration baseline under varying room lighting.
- 2

Debounce & Rapid Motion Test
Pass hands rapidly across strings to confirm zero missed or double triggers.
- 3

Crosstalk & Multi-Beam Test
Confirm adjacent laser beams do not trigger neighboring sensors.
- 4

Live Presentation Test
Pre-calibrate sensor thresholds under actual presentation room light.

Measurable Target Metrics

- Trigger Response Latency**
Under 100 ms from beam interruption to audible tone.
- Missed / Double Triggers**
Zero false negatives or duplicate notes during rapid-motion testing.
- Beam Crosstalk**
No cross-triggering between adjacent sensor channels.
- Calibration Consistency**
Stable thresholds re-established under presentation-room lighting.

Feature Grading Scope Breakdown

ESSENTIAL CORE

ADC beam detection, dynamic thresholding, clean tone generation on the speaker, and synchronized LED indicators.

OPTIONAL EXTENSIONS

ESP32 MP3 audio playback, WS2812 rainbow lighting animations, and Bluetooth MIDI expansion.

Challenges & Future Scope

Foreseen Engineering Challenges

1 Ambient Light Interference

Fluctuating room lighting shifts LDR resistance; mitigated by auto-calibrating the ADC baseline at startup.

2 Mechanical Beam Alignment

5mW lasers require precise optical targeting onto small LDR surfaces; addressed via rigid frame construction.

3 Debounce Algorithm Tuning

Distinguishing quick hand pass-throughs from deliberate note holds requires fine-tuned hysteresis timing.

Post-course potential enhancements beyond initial proposal scope.

Future Expansion Ideas

• Polyphonic Chord Playback

Hardware timer mixing or DAC integration to play multiple simultaneous notes when several beams break together.

• Full Octave Expansion

Expand the sensor array to 12+ notes using external analog multiplexers (e.g., CD4051) on the ATmega32 ADC ports.

• On-the-Fly Sensitivity Tuning

Add a rotary encoder / potentiometer interface to dynamically adjust trigger sensitivity without reflashing.

THANK YOU

Questions & Feedback Are Welcome

Laser Harp: Touchless Musical Instrument | ATmega32 Embedded Microcontroller Project

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References & Image Sources

ATmega32 Component Image

<https://www.ic-components.com/upfile/images/3c/20260304223054833.jpg>

Source: www.ic-components.com